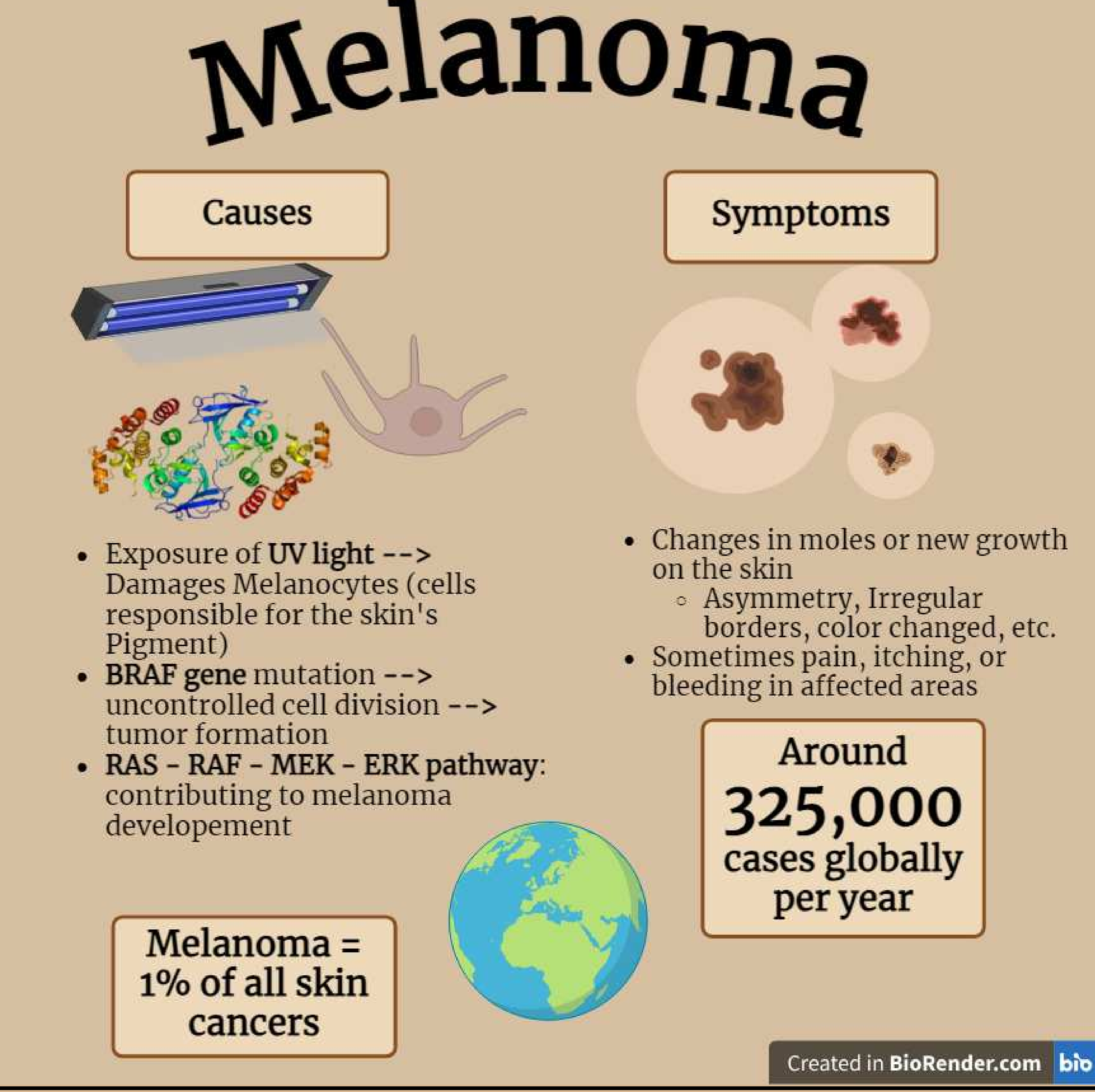
An Investigation of the Effect of Astragalus membranaceus on BRAF expression and cell viability using B16-F10 Cells



Research Question

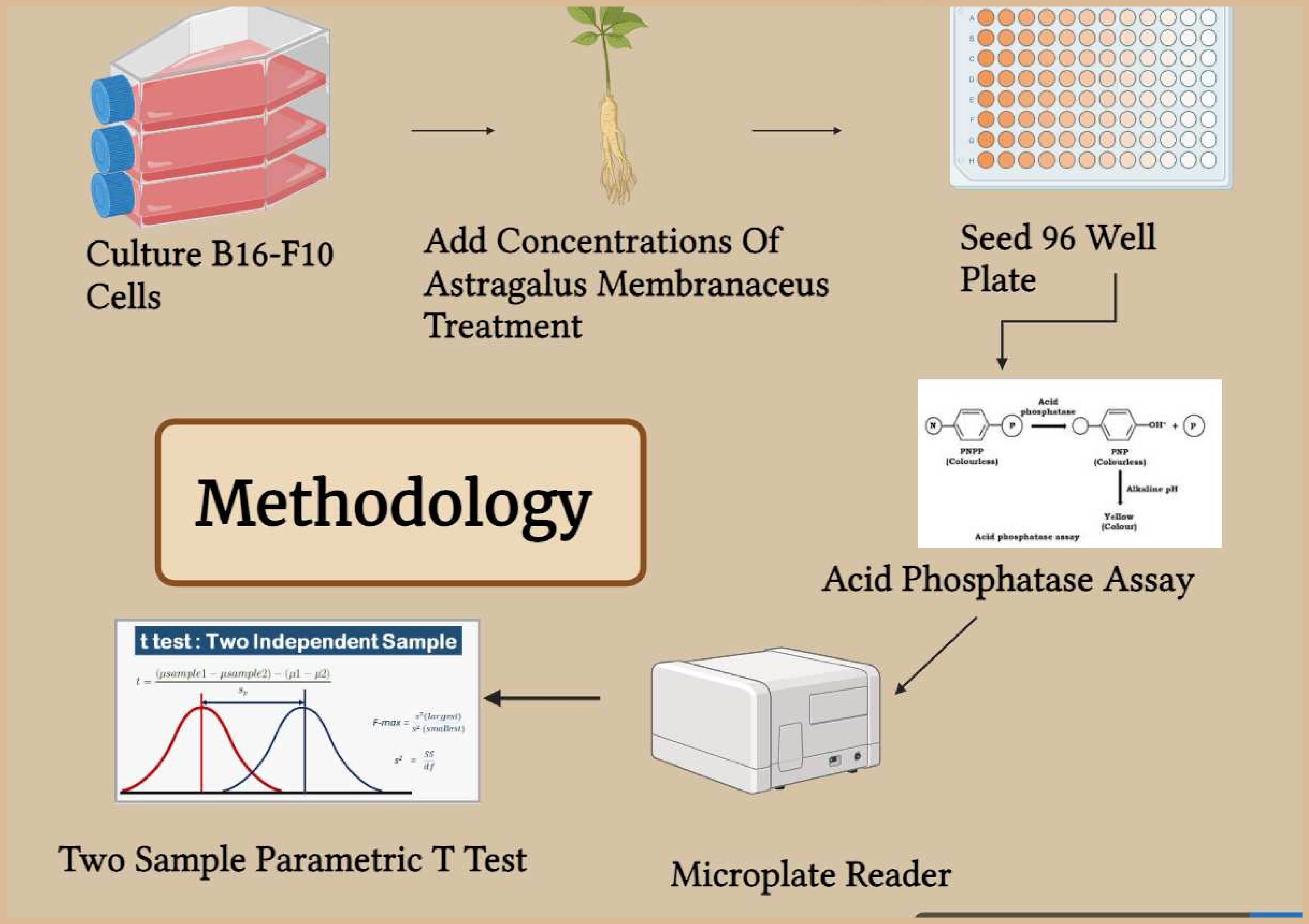
How do varying concentrations of Astragalus Membranaceus affect the cell viability of B16-F10 cells?

Hypothesis

<u>If B16-F10 cells are given variousconcentrations of Astragalus membranaceus then cell viability will be reduced.</u> Astragalus membranaceus showed to inhibit the ERK pathway which is part of a larger MAPK cell signaling pathway. This is significant because the mutated BRAF gene codes the B-raf protein which plays a big role in continued activation of the MAPK cell signaling pathway. Therefore it can be expected that high levels of Astragalus membranaceus should inhibit the MAPK pathway resulting in the reduction of cell proliferation.

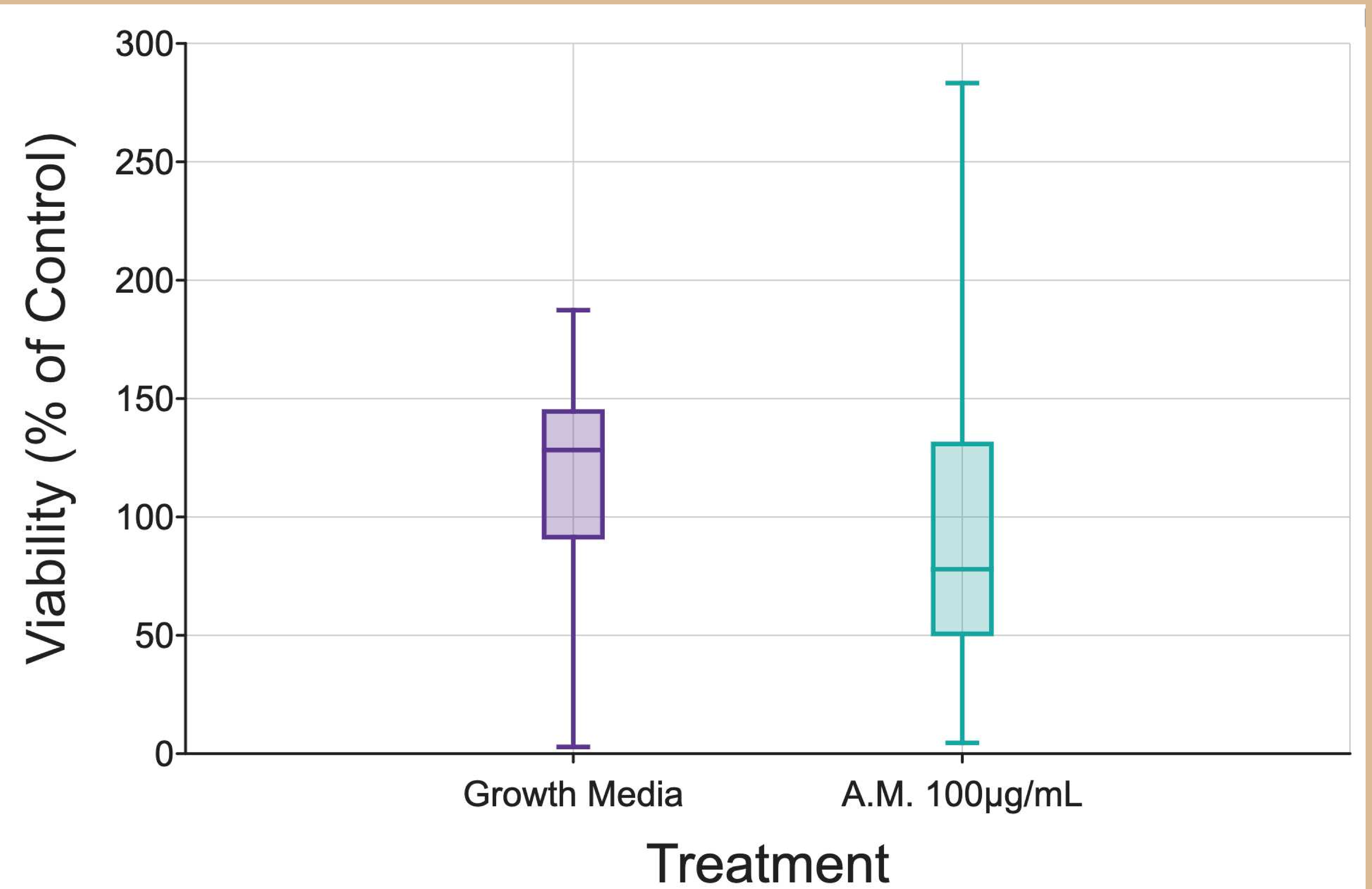
IV: ASTRAGALUS MEMBRANACEOUS
DV: % CELL VIABILITY
NEGATIVE CONTROL: GROWTH MEDIA (90 ML DMEM AND 10% FBS)

Methodology



Data

Figure 1: Astragalus Membranaceous Treatment on cell viability as % of control

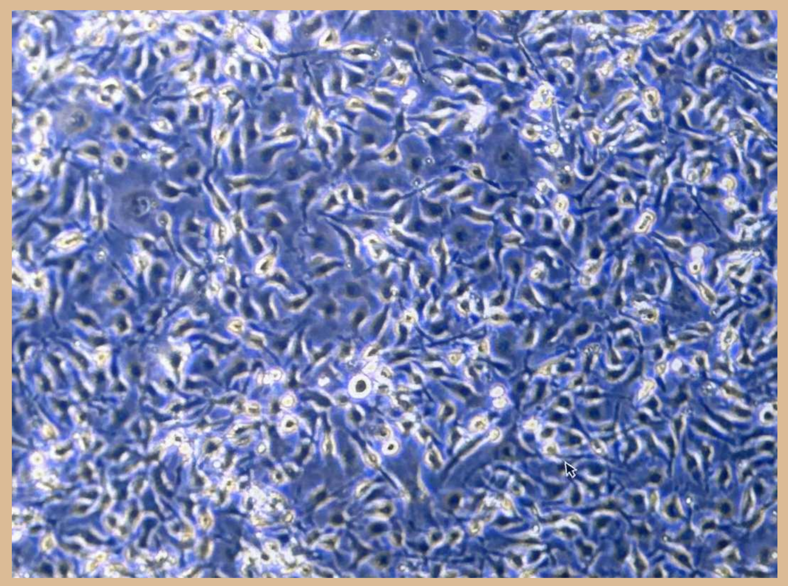


Note: Box and whiskers plot shows the median value (line), interquartile range (box), and the extent of the data (whiskers above and below at min / max data points).

Discussion

In Accordance to Figure 1, 100 µg/mL of Astragalus Membranaceous showed a reduction in the average cell viability of B16-F10 Cells, when being compared to the negative control: Growth Media. By comparing the two groups in % viability of control, Astragalus membranaceous showed a lower median viability than growth media, as indicated by the box and whisker plot. Treatment with Astragalus Membranaceous resulted in widespread variability of the data leading to a p value of 0.20 indicating that there is no significant difference between the groups. This statistical difference, goes against the conclusion that the reduction in cell viability was not due to random chance and was impacted by the mechanisms of Astragalus Membranaceous. Although further replicates may indicate a significant change in the, metabolic function of Astragalus Membranaceous and should be researched further.

Figure 2: Cultured B16:F10 Cells under 10x Microscope



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